

Exploring Performance–Efficiency Trade-offs in Semantic Segmentation on the Cityscapes Dataset

Yusuf Berkan Gokce
Eindhoven University of Technology
Artificial Intelligence and Engineering Systems
Eindhoven, Netherlands
y.b.gokce@student.tue.nl

Abstract—Semantic segmentation requires assigning a class label to each pixel in an image and is a fundamental task in computer vision for detailed scene understanding. While modern deep learning architectures achieve strong segmentation performance, they often incur substantial computational cost, creating a trade-off between accuracy and efficiency. In this work, we investigate this trade-off on the Cityscapes dataset by evaluating a diverse set of semantic segmentation architectures. U-Net is used as a baseline model, while DeepLabV3 and DeepLabV3+ variants with ResNet backbones are evaluated for peak performance. ENet, BiSeNetV2, and DeepLabV3+ with MobileNetV3 are evaluated as efficiency-oriented architectures. Models are compared using segmentation metrics, including Mean Dice Score and Mean Intersection-over-Union (Mean IoU), as well as efficiency metrics such as GFLOPs, model size, inference speed, and an Efficiency Score. The results show that DeepLabV3 with ResNet101 achieves the highest segmentation performance, whereas ENet achieves the highest efficiency score with substantially lower computational requirements. DeepLabV3+ with MobileNetV3 and BiSeNetV2 provide intermediate trade-offs between these extremes. These findings demonstrate that no single architecture is optimal across all evaluation criteria and highlight the importance of selecting models according to the performance and resource constraints of the target application.

Index Terms—semantic segmentation, Cityscapes dataset, DeepLabV3, U-Net, ENet, BiSeNetV2, computational efficiency, real-time inference, neural network architectures, performance–efficiency trade-off

I. INTRODUCTION

Semantic segmentation is a core task in computer vision where each pixel in an image is assigned a class label. Unlike image classification, which gives one label per image, segmentation provides detailed information about the scene. This is especially important in applications such as autonomous driving, where it is necessary to distinguish between roads, pedestrians, vehicles, and other objects.

Although deep learning has improved segmentation performance significantly, the task is still challenging. Datasets like Cityscapes contain complex urban scenes with occlusions, varying object sizes, and class imbalance, making accurate prediction difficult. Earlier work such as Fully Convolutional Networks (FCNs) introduced end-to-end pixel-wise prediction [1], while later approaches such as DeepLab improved segmentation quality through atrous convolutions and multi-scale context aggregation [2]. Encoder–decoder architectures such as U-Net further demonstrated the importance of combining

semantic and spatial information through skip connections [4]. More recently, lightweight architectures including ENet [3] and BiSeNetV2 [8] have been proposed to enable real-time segmentation with reduced computational requirements.

Despite these advances, a fundamental trade-off remains between segmentation accuracy and computational efficiency. High-capacity architectures often achieve superior performance but require greater computational resources, while lightweight models offer faster inference and lower complexity at the cost of reduced segmentation quality. Understanding this trade-off is important when selecting models for practical deployment scenarios with limited computational resources.

In this work, we investigate this trade-off on the Cityscapes dataset by evaluating a diverse set of semantic segmentation architectures, including U-Net, DeepLabV3, DeepLabV3+, ENet, and BiSeNetV2 variants. These models span a wide range of architectural complexity and computational requirements, enabling a comparison of performance and efficiency characteristics. The following research question guides this study:

What is the trade-off between segmentation accuracy and computational efficiency across different semantic segmentation architectures on the Cityscapes dataset?

To answer this question, we compare the models using segmentation and efficiency metrics, including Dice score, Intersection-over-Union (IoU), computational cost (GFLOPs), model size, and inference speed (FPS). Through this analysis, we aim to provide insights into how different model choices affect the balance between accuracy and efficiency.

II. METHODOLOGY

A. Dataset

All models are trained and evaluated on the Cityscapes dataset, which contains high-resolution urban street images with pixel-level annotations across 19 semantic classes such as roads, buildings, pedestrians, and vehicles [6]. The dataset presents realistic challenges including scale variation, occlusions, and class imbalance. In this project, the dataset was already split into training and validation sets. The test set was not directly accessible during development and evaluation; instead, it was hosted on an external submission server. Final

model performance on the test set was therefore obtained only through submission, ensuring a fair and unbiased evaluation.

B. Baseline Model

For both benchmarks, U-Net is used as the baseline architecture. U-Net follows a symmetric encoder–decoder structure in which the encoder progressively extracts higher-level semantic features through convolution and pooling operations, while the decoder restores spatial resolution through upsampling [4]. A key component of U-Net is the use of skip connections between corresponding encoder and decoder stages. These connections preserve fine-grained spatial information that would otherwise be lost during downsampling, improving localization accuracy and boundary reconstruction [4]. U-Net is chosen as a baseline in this study due to its simplicity, relatively low implementation complexity, and strong performance for segmentation tasks. It provides a reliable reference point for both accuracy and efficiency, making it suitable for evaluating improvements introduced by more advanced architectures.

C. Peak Performance Models

1) *DeepLabV3 with ResNet50*: To improve segmentation performance beyond the U-Net baseline, DeepLabV3 with a ResNet50 backbone is evaluated. DeepLabV3 employs atrous convolutions and an Atrous Spatial Pyramid Pooling (ASPP) module to capture contextual information at multiple scales while maintaining feature-map resolution [2]. The ResNet50 backbone serves as a pretrained feature extractor and provides rich hierarchical representations that are subsequently fine-tuned on the Cityscapes dataset. Compared to U-Net, DeepLabV3 benefits from larger receptive fields and improved multi-scale context modeling, making it a strong candidate for high-performance semantic segmentation [5].

2) *DeepLabV3 with ResNet101*: To further increase model capacity, the ResNet50 backbone is replaced by ResNet101. The deeper backbone contains substantially more residual bottleneck blocks, allowing the network to learn more complex feature representations [7]. The additional depth improves contextual understanding and fine-detail representation, particularly in challenging urban scenes. However, these gains come at the cost of increased computational complexity and model size. This model represents the highest-capacity DeepLabV3 configuration evaluated in the peak-performance benchmark.

3) *DeepLabV3+ with ResNet101*: DeepLabV3+ extends DeepLabV3 by introducing a decoder module that refines segmentation boundaries using low-level spatial features [9]. While DeepLabV3 primarily focuses on contextual information through ASPP, DeepLabV3+ additionally improves object boundary reconstruction. The ResNet101 backbone provides high-capacity feature extraction, while the decoder helps recover spatial details lost during feature encoding [7]. As a result, DeepLabV3+ aims to achieve stronger segmentation accuracy, particularly around object boundaries and small structures.

D. Efficiency-Oriented Models

1) *DeepLabV3+ with MobileNetV3*: To investigate the impact of efficient backbone networks, DeepLabV3+ is also evaluated using a MobileNetV3 backbone. MobileNetV3 is specifically designed for resource-constrained environments and achieves efficiency through lightweight bottleneck blocks, depthwise separable convolutions, and channel attention mechanisms [9]. Combined with the DeepLabV3+ decoder, this architecture aims to preserve much of the segmentation capability of larger DeepLab models while significantly reducing computational cost and improving inference speed.

2) *BiSeNetV2*: BiSeNetV2 is a real-time semantic segmentation architecture that separates spatial detail extraction and semantic feature extraction into two dedicated branches [8]. The Detail Branch preserves high-resolution spatial information, while the Semantic Branch focuses on efficient contextual representation learning. The outputs of both branches are fused using a Bilateral Guided Aggregation (BGA) module, enabling the network to combine detailed spatial information with high-level semantic understanding [8]. This design allows BiSeNetV2 to achieve a favorable balance between segmentation accuracy and computational efficiency, making it well suited for real-time applications.

3) *ENet*: ENet is a lightweight semantic segmentation architecture designed for real-time applications. Unlike traditional segmentation networks, ENet reduces computational cost through early downsampling, bottleneck modules, asymmetric convolutions, and dilated convolutions [3]. Another important design choice is the use of a relatively large encoder and a lightweight decoder. As a result, most computation is concentrated in feature extraction while the reconstruction stage remains efficient. These characteristics enable ENet to achieve low model size and computational complexity while maintaining reasonable segmentation performance.

E. Evaluation Metrics

For the peak-performance benchmark, the primary evaluation metrics are Mean Dice Score and Mean Intersection-over-Union (Mean IoU):

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (1)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (2)$$

To evaluate computational efficiency, additional metrics including computational complexity (GFLOPs), model size, and inference speed measured in frames per second (FPS) are considered. Following the efficiency benchmark, an efficiency score is defined as:

$$Efficiency\ Score = \frac{Dice}{TFLOPs} \quad (3)$$

This metric provides a normalized measure of segmentation performance relative to computational cost, allowing direct comparison between high-capacity and lightweight architectures.

TABLE I
PEAK PERFORMANCE COMPARISON ON THE CITYSCAPES TEST SET

Model	Mean Dice	Mean IoU	Flat	Cons.	Obj.	Nat.	Sky	Human	Veh.
U-Net (Baseline)	0.4755	0.3882	0.6027	0.7907	0.1883	0.1533	0.0870	0.1766	0.2776
DeepLabV3 (ResNet50)	0.5506	0.4458	0.6626	0.7853	0.3000	0.3474	0.0812	0.3468	0.3826
DeepLabV3Plus (ResNet101)	0.5546	0.4503	0.6712	0.7712	0.3106	0.3302	0.0677	0.3252	0.3897
DeepLabV3 (ResNet101)	0.5600	0.4569	0.6749	0.8092	0.3265	0.3653	0.0666	0.3546	0.3474

TABLE II
EFFICIENCY COMPARISON ON THE CITYSCAPES TEST SET

Model	Dice/TFLOPs	Size (MB)	GFLOPs	FPS	Mean Dice	Mean IoU
U-Net (Baseline)	0.1855	65.9340	2563.5586	8.8600	0.4755	0.3882
BiSeNetV2	2.1734	14.6114	232.6492	28.2100	0.5056	0.4097
DeepLabV3+ (MobileNetV3)	4.2784	26.7672	120.0131	72.2800	0.5135	0.4137
ENet	15.4767	1.6032	29.0487	46.9100	0.4496	0.3686

III. RESULTS

This section presents the quantitative and qualitative results of the evaluated models on the Cityscapes test set. All results were obtained through submission to the evaluation server.

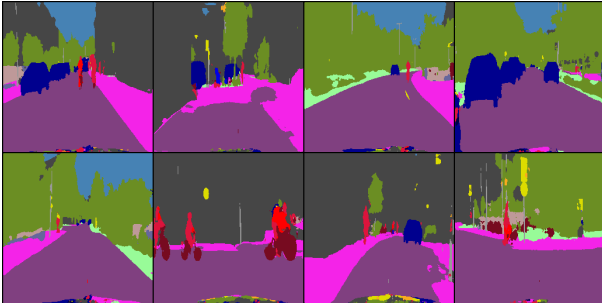


Fig. 1. Segmentation outputs of the U-Net baseline model.

Table I summarizes the quantitative results for the peak-performance benchmark. Among the evaluated models, DeepLabV3 with ResNet101 achieved the highest Mean Dice score (0.5600) and Mean IoU (0.4569). DeepLabV3Plus with ResNet101 achieved a Mean Dice score of 0.5546 and a Mean IoU of 0.4503, while DeepLabV3 with ResNet50 achieved a Mean Dice score of 0.5506 and a Mean IoU of 0.4458. The U-Net baseline achieved a Mean Dice score of 0.4755 and a Mean IoU of 0.3882. Class-wise Dice scores for the seven Cityscapes categories are also reported in Table I.

Table II presents the efficiency benchmark results. ENet achieved the highest Efficiency Score (15.4767) and had the smallest model size (1.6032 MB). DeepLabV3Plus with MobileNetV3 achieved a Mean Dice score of 0.5135 and a Mean IoU of 0.4137, while BiSeNetV2 achieved a Mean Dice score of 0.5056 and a Mean IoU of 0.4097. The U-Net baseline achieved a Mean Dice score of 0.4755 and a Mean IoU of 0.3882, with a model size of 65.9340 MB and computational complexity of 2563.5586 GFLOPs.

Figure 1 shows representative segmentation outputs of the U-Net baseline model. These predictions provide a quali-

tative reference for comparison with the peak-performance and efficiency-oriented architectures evaluated in this study. On the other hand, Figures 2 and 3 show representative segmentation outputs generated by the best performing peak-performance and efficiency-oriented architectures on Cityscapes test images.

IV. DISCUSSION

The results provide a clear answer to the research question: there is a strong trade-off between segmentation accuracy and computational efficiency across semantic segmentation architectures. Models designed for maximum segmentation performance consistently achieved higher Dice and IoU scores, whereas architectures optimized for efficiency achieved substantially lower computational cost and model size.

Among the peak-performance models, DeepLabV3 with ResNet101 achieved the highest Mean Dice score (0.5600) and Mean IoU (0.4569). This result is consistent with the literature, where DeepLab-based architectures are known to benefit from atrous convolutions and multi-scale context aggregation through the ASPP module [2]. The deeper ResNet101 backbone also provides stronger feature representations than ResNet50, which is reflected in the performance improvement observed in Table I. Interestingly, DeepLabV3 and DeepLabV3Plus with a ResNet101 backbone achieved very similar results, with DeepLabV3 performing slightly better in our experiments. Although DeepLabV3Plus was proposed as an extension of DeepLabV3 [9], the observed performance gap was minimal and may be attributed to differences in training dynamics, hyperparameter settings, or optimization. This suggests that both architectures are capable of achieving comparable levels of segmentation performance on the Cityscapes dataset, and that further improvements may depend more on training and tuning strategies than on architectural modifications alone.

The efficiency-oriented benchmark further highlights the trade-off between performance and computational cost. ENet achieved the highest Efficiency Score (15.4767) while requiring only 29.0 GFLOPs and 1.6 MB of storage. This aligns with

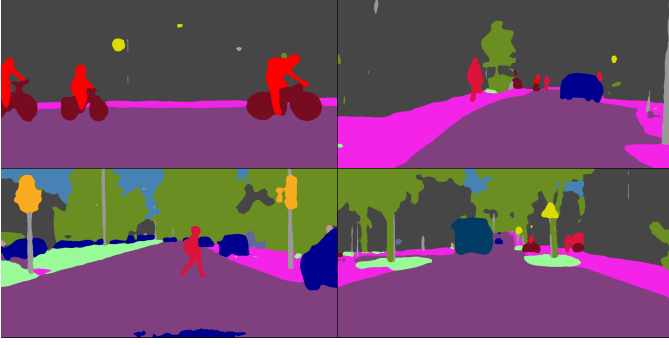


Fig. 2. Segmentation outputs of DeepLabV3 with ResNet101.

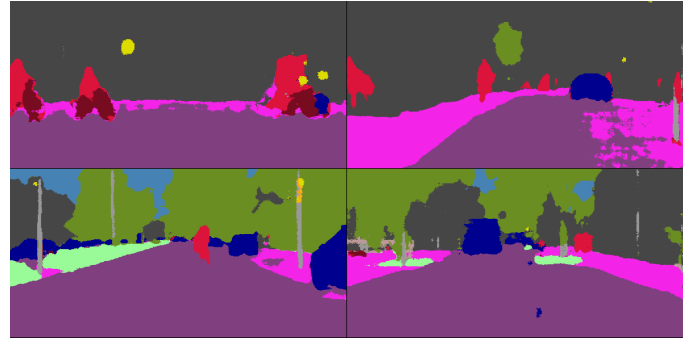


Fig. 3. Segmentation outputs of ENet.

the original ENet design philosophy, which prioritizes real-time deployment through aggressive parameter reduction and lightweight bottleneck modules [3]. However, this efficiency comes at the cost of segmentation quality, as ENet achieved the lowest Dice and IoU scores among the evaluated architectures.

The inclusion of DeepLabV3Plus with MobileNetV3 and BiSeNetV2 provides additional insight into the middle ground between performance and efficiency. DeepLabV3Plus with MobileNetV3 achieved the highest segmentation accuracy among the efficiency-oriented models while maintaining substantially lower computational requirements than the ResNet-based DeepLab variants. BiSeNetV2 achieved slightly lower segmentation performance but offered a smaller model size and lower computational complexity. These results are consistent with the design goals of both architectures, which aim to balance accuracy and efficiency for practical deployment scenarios [8], [9].

A notable observation is that the highest-performing model and the most efficient model differ substantially. DeepLabV3 with ResNet101 achieved the best segmentation performance but required significantly greater computational resources, whereas ENet achieved the best Efficiency Score while sacrificing segmentation accuracy. This demonstrates that no single architecture dominates across all evaluation criteria. Instead, model selection depends on application requirements. For example, autonomous driving systems deployed on powerful hardware may prioritize segmentation accuracy, whereas embedded or edge devices may benefit more from lightweight architectures.

Several limitations should be considered when interpreting these results. First, only a limited set of architectures and backbone configurations was evaluated. Other modern architectures, including transformer-based segmentation models, may exhibit different accuracy–efficiency trade-offs. Second, all experiments were conducted on the Cityscapes dataset, which focuses on urban driving scenes. As a result, the conclusions may not generalize to other domains such as medical imaging or aerial imagery. Finally, the comparison is influenced by the specific training procedures and hyperparameter settings used in this study, which may affect the relative ranking of the evaluated models.

V. CONCLUSION

This study investigated the trade-off between segmentation accuracy and computational efficiency across different semantic segmentation architectures on the Cityscapes dataset. DeepLabV3 with ResNet101 achieved the highest segmentation performance, while ENet provided the greatest computational efficiency through its low model size and computational cost. BiSeNetV2 and DeepLabV3+ with MobileNetV3 offered intermediate solutions that balanced accuracy and efficiency. These results confirm that no single architecture is optimal across all evaluation criteria, highlighting the importance of selecting models according to deployment requirements. Future work could extend this analysis to transformer-based segmentation architectures and model compression techniques to further improve the accuracy–efficiency trade-off.

VI. CODE AVAILABILITY

The implementation, trained segmentation models, and experimental code used in this work are publicly available at: <https://github.com/BerkanGokce/NNCV.git>

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